MAE 143B HW8

6.2 **(a)** Calculate the magnitude and phase of

$$G(s) = \frac{1}{s+10}$$

by hand for $\omega = 1, 2, 5, 10, 20, 50,$ and 100 rad/sec.

(b) Sketch the asymptotes for $G(s)$ according to the Bode plot rules, and compare these with your computed results from part (a).

6.4 *Real poles and zeros.* Sketch the asymptotes of the Bode plot magnitude and phase for each of the listed open-loop transfer functions. After completing the hand sketches, verify your result using MATLAB. Turn in your hand sketches and the MATLAB results on the same scales.

(a) $L(s) = \dfrac{1}{s(s+1)(s+5)(s+10)}$

(b) $L(s) = \dfrac{(s+2)}{s(s+1)(s+5)(s+10)}$

(c) $L(s) = \dfrac{(s+2)(s+6)}{s(s+1)(s+5)(s+10)}$

(d) $L(s) = \dfrac{(s+2)(s+4)}{\cdots}$

6.5 *Complex poles and zeros.* Sketch the asymptotes of the Bode plot magnitude and phase for each of the listed open-loop transfer functions, and approximate the transition at the second-order break point, based on the value of the damping ratio. After completing the hand sketches, verify your result using MATLAB. Turn in your hand sketches and the MATLAB results on the same scales.

(a) $L(s) = \dfrac{1}{s^2+3s+10}$

(b) $L(s) = \dfrac{1}{s(s^2+3s+10)}$

(c) $L(s) = \dfrac{(s^2+2s+8)}{s(s^2+2s+10)}$

(d) $L(s) = \dfrac{(s^2+2s+12)}{s(s^2+2s+10)}$

(e) $L(s) = \dfrac{(s^2+1)}{s(s^2+4)}$

(f) $L(s) = \dfrac{(s^2+4)}{s(s^2+1)}$

6.9 A certain system is represented by the asymptotic Bode diagram shown in Fig. 6.87. Find and sketch the response of this system to a unit-step input (assuming zero initial conditions).

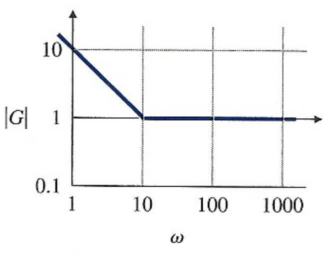

Figure 6.87
Magnitude portion of Bode plot for Problem 6.9